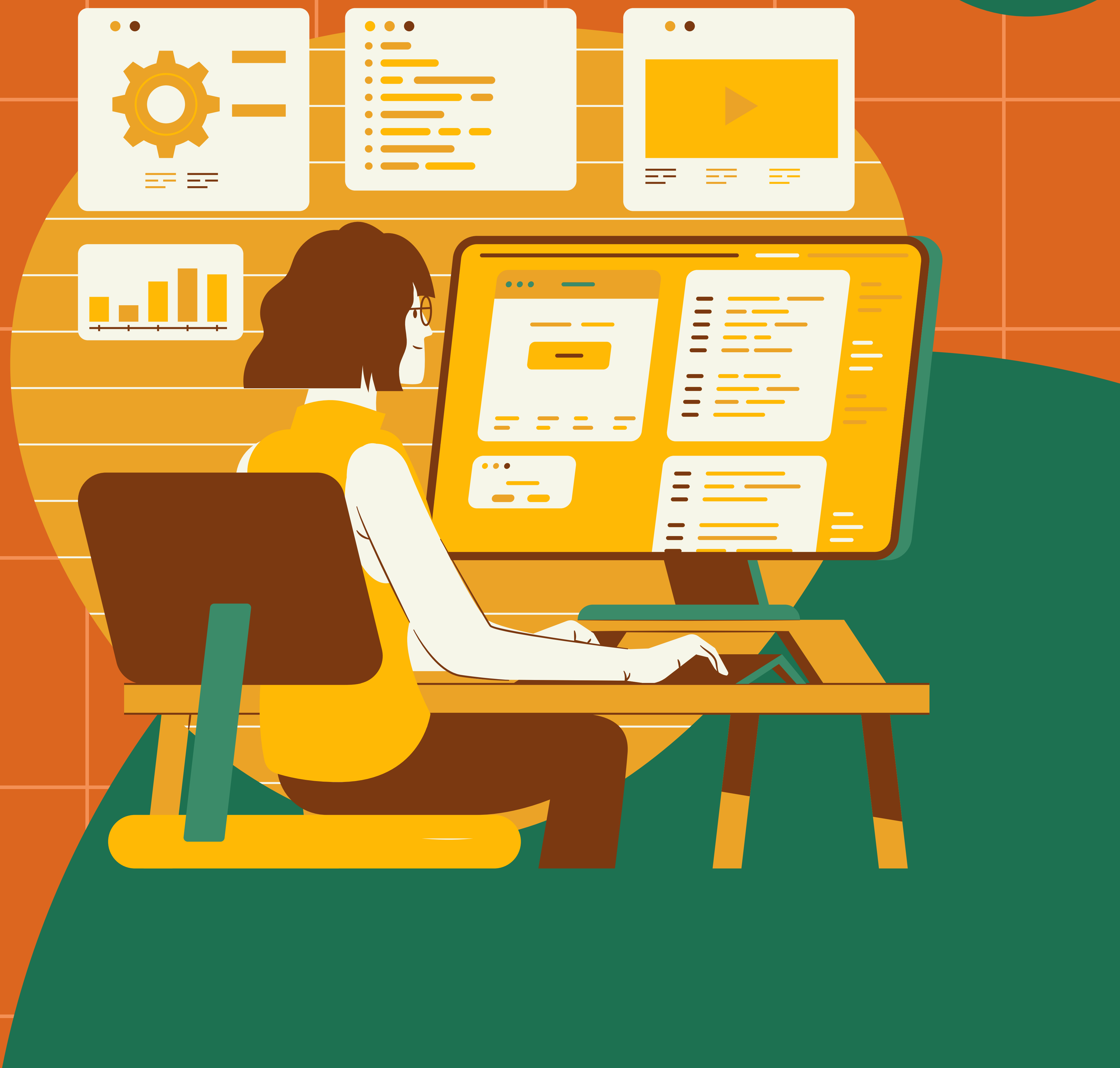
INTRODUCTION TO COMPUTERS APPRECIATION OF COMPUTING IN DIFFERENT FIELD

MODULE 1: Fundamentals of Computing

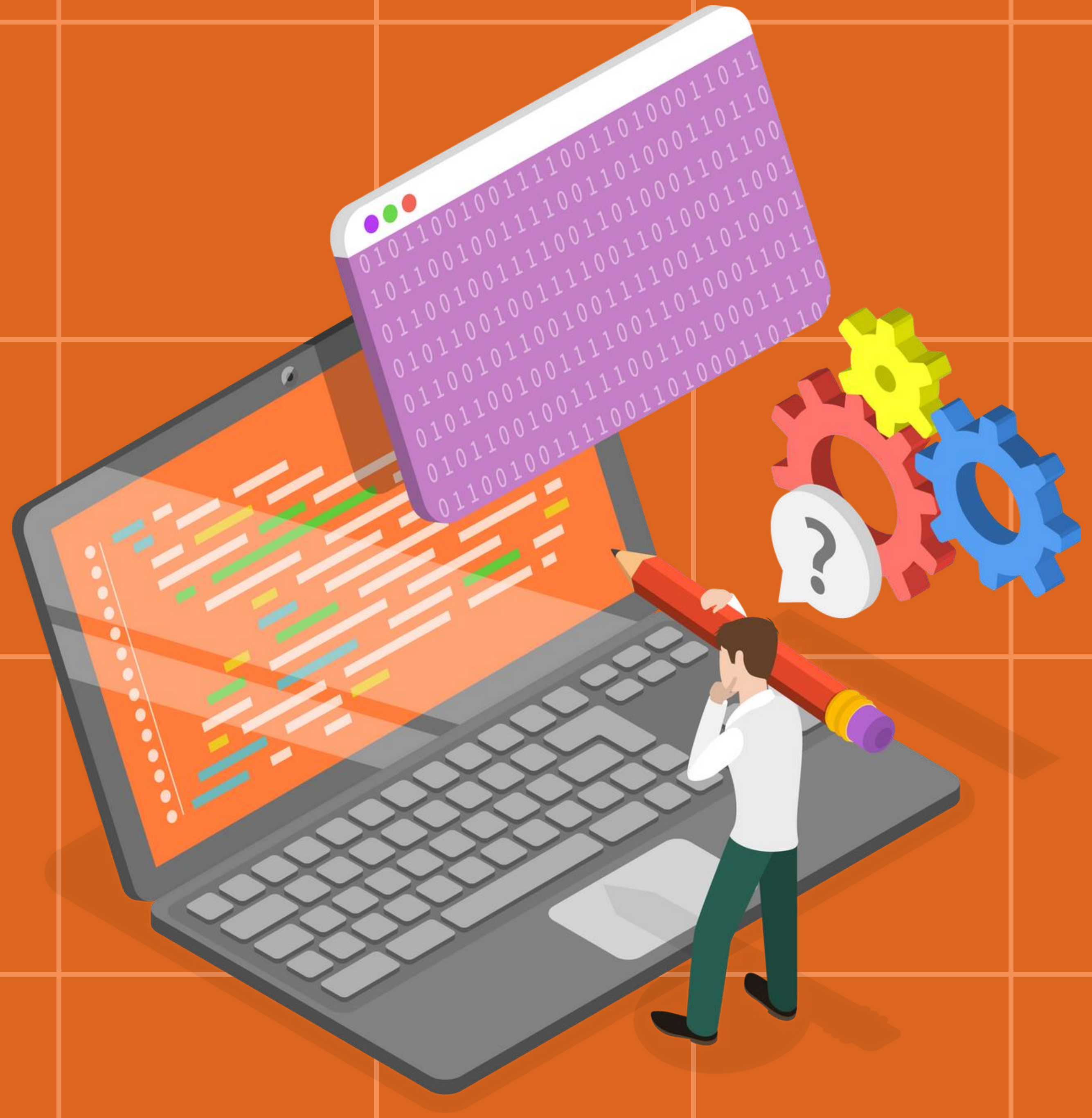
CCC100



COMPUTING

Computing is the process of utilizing computer technology to complete a task. Computing may involve computer hardware and/or software, but must involve some form of a computer system. Most individuals use some form of computing every day whether they realize it or not.

The ACM Computing Curricula 2005 defined computing as:



In a *general way*, we can define *computing* to mean any goal-oriented activity requiring, benefiting from, or creating computers. Thus, computing includes designing and building hardware and software systems for a wide range of purposes; *processing*, *structuring*, and *managing various kinds of information*; *doing scientific studies using computers*; *making computer systems behave intelligently*; *creating and using communications and entertainment media*; *finding and gathering information relevant to any particular purpose*, and *so on*. The list is virtually endless, and the possibilities are vast.[1]

DEFINITION OF COMPUTING

Defines five sub-disciplines of the computing field:

- Computer Science
- Computer Engineering
- Information Systems
- Information Technology
- Software Engineering

The term "computing" is also synonymous with counting and calculating.

COMPUTER ENGINEERING

- Is concerned with the design and construction of computers and computer-based systems.
- It involves the study of hardware, software, communications, and the interaction among them.
- Its curriculum focuses on the theories, principles, and practices of traditional electrical engineering and mathematics and applies them to the problems of designing computers and computer-based devices.

COMPUTER SCIENCE

Spans a wide range, from its theoretical and algorithmic foundations to cutting-edge developments in robotics, computer vision, intelligent systems, bioinformatics, and other exciting areas. We can think of the work of computer scientists as falling into three categories.

- ✓ **They design and implement software**
- ✓ **They devise new ways to use computers**
- ✓ **They develop effective ways to solve computing problems**

INFORMATION SYSTEMS

- Specialists focus on integrating information technology solutions and business processes to meet the information needs of businesses and other enterprises, enabling them to achieve their objectives in an effective, efficient way.

- This discipline's perspective on information technology emphasizes information, and views technology as an instrument for generating, processing, and distributing information.

- They must understand both technical and organizational factors, and they must be able to help an organization determine how information and technology-enabled business processes can provide a competitive advantage

- The *information systems* specialist plays a key role in determining the requirements for an organization's information systems and is active in their specification, design, and implementation.

INFORMATION TECHNOLOGY

- Is a label that has two meanings. In the broadest sense, the term information technology is often used to refer to all of computing. In academia, it refers to undergraduate degree programs that prepare students to meet the computer technology needs of business, government, healthcare, schools, and other kinds of organizations. In some nations, other names are used for such degree programs.

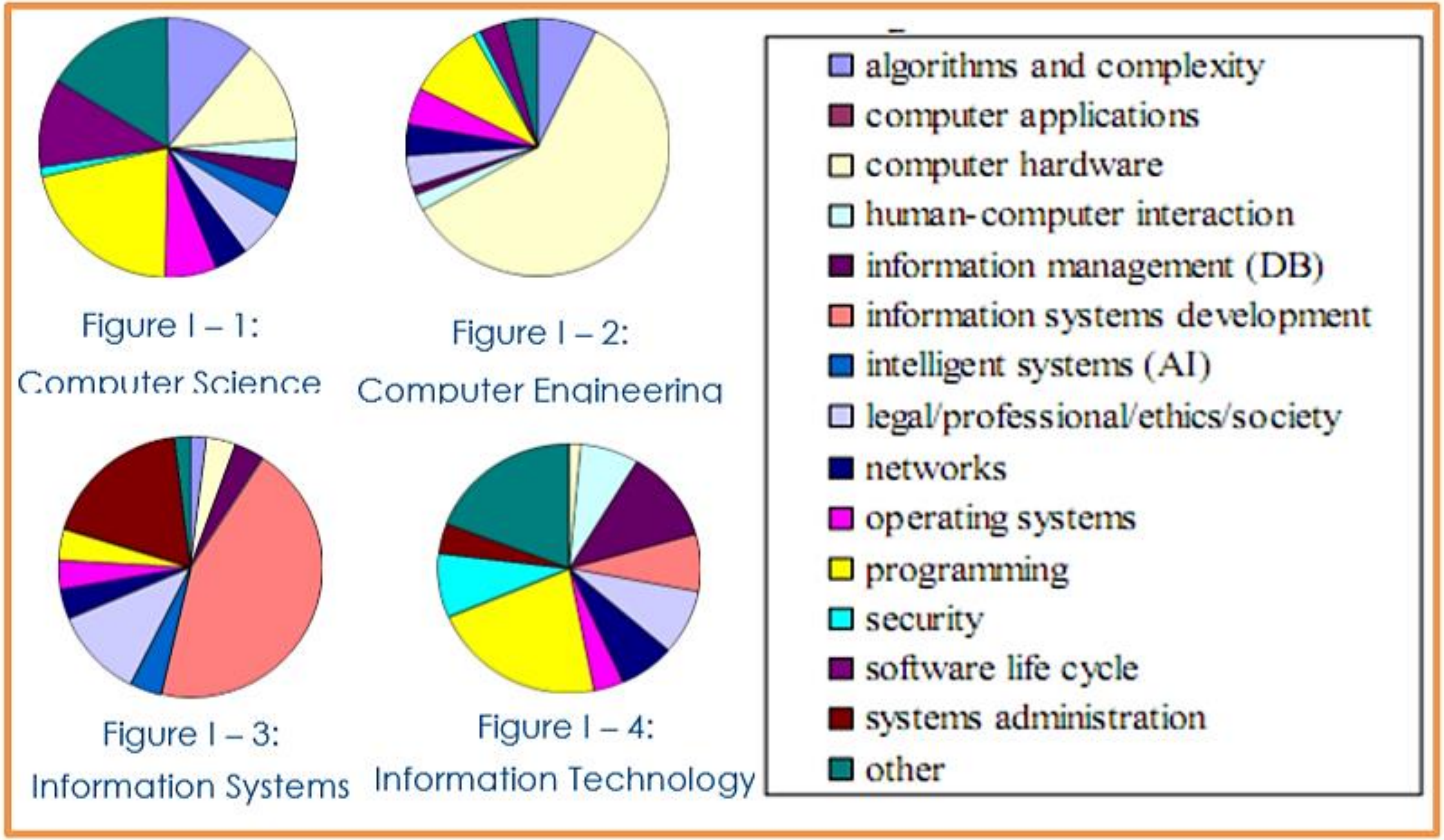
- In the previous section, we said that Information Systems focuses on the information aspects of information technology. Information Technology is the complement of that perspective: its emphasis is on the technology itself more than on the information it conveys.

SOFTWARE ENGINEERING

- Software engineering is the discipline of developing and maintaining software systems that behave reliably and efficiently, are affordable to develop and maintain, and satisfy all the requirements that Customers have defined for them. More recently, it has evolved in response to factors such as the growing impact of large and expensive software systems in a wide range of situations and the increased importance of software in safety-critical applications.

- Software engineering is different in character from other engineering disciplines due to both the intangible nature of software and the discontinuous nature of software operation. It seeks to integrate the principles of mathematics and computer science with the engineering practices developed for tangible, physical artifacts.

KNOWLEDGE AREAS: DIFFERENCES



KNOWLEDGE AREAS: COMMONALITIES

- ✓ Computer foundational topics
- ✓ Computer programming
- ✓ Capabilities and limitations of computers
- ✓ Software lifecycle issues
- ✓ Processes, both computing and professional
- ✓ Advanced computing topics
- ✓ Professionalism
- ✓ Applications to join theory and skills
- ✓ Capstone projects/Thesis

NATURE OF THE FIELD OF STUDY

Bachelor of Science in Computer Science (BSCS)

The BS Computer Science program includes the study of computing concepts and theories, algorithmic foundations and new developments in computing. The program prepares students to design and create algorithmically complex software and develop new and effective algorithms for solving computing problems.

The program also includes the study of the standards and practices in Software Engineering. It prepares students to acquire skills and disciplines required for designing, writing and modifying software components, modules and applications that comprise software solutions.

Primary Job Roles

- Software Engineer
- Systems Software Developer
- Research and Development computing professional
- Applications Software Developer
- Computer Programmer

Secondary Job Roles

- System Analyst
- Data Analyst
- Quality Assurance Specialist
- Software Support Specialist

NATURE OF THE FIELD OF STUDY

Bachelor of Science in Information Systems (BSIS)

The BS Information Systems Program includes the study of application and effect of information technology to organizations. Graduates of the program should be able to implement an information system, which considers complex technological and organizational factors affecting it. These include components, tools, techniques, strategies, methodologies, etc.

Graduates are able to help an organization determine how information and technology-enabled business processes can be used as strategic tool to achieve a competitive advantage.

Primary Job Roles

- Organizational Process Analyst
- Data Analyst
- Solutions Specialist
- System Analyst
- IS Project Management Personnel

Secondary Job Roles

- Applications Developer
- End User Trainer
- Documentation Specialist
- Quality Assurance Specialist

NATURE OF THE FIELD OF STUDY

Bachelor of Science in Information Technology (BSIT)

The BS Information Technology includes the study of the utilization of both hardware and software technologies involving planning, installing, customizing, operating, managing and administering, and maintaining information technology infrastructure that provides computing solutions to address the needs of an organization.

The program prepares graduates to address various user needs involving the selection, development, application, integration and management of computing technologies within an organization.

Primary Job Roles

- Web and Applications Developer
- Junior Database Administrator
- Systems Administrator
- Network Engineer
- Junior Information Security Administrator
- Systems Integration Personnel
- IT Audit Assistant
- Technical Support Specialist

Secondary Job Roles

- QA Specialist
- System Analyst
- Computer Programmer

COMPONENTS OF THE SYSTEM UNIT, INPUT, OUTPUT, AND STORAGE

MODULE 2: Fundamentals of Computing

CCC100

DIFINITION OF COMPUTERS

A computer is an electronic device, operating under the control of instructions stored in its own memory that can accept data, process the data according to specified rules, produce results and store the results for future use.

There are four basic operations involved:

INPUT	Feeding data to the computer from the external environment.
PROCESS	Manipulating and transforming of data by arithmetic and logical instructions
STORAGE	Recording or storing of data involved in all operations.
OUTPUT	Producing data by presenting results derived from processing.

CHARACTERISTICS OF COMPUTER

SPEED	The computer can process data very fast, at the rate of millions of instructions per second.
ACCURACY	Computer provides a high degree of accuracy.
DILIGENCE	When used for a longer period of time, the computer does not get tired or fatigued.
VERSATILITY	Computer is capable of performing almost any task, if the task can be reduced to a finite series of logical steps.
STORAGE CAPABILITY	Large volumes of data and information can be stored in the computer and also retrieved whenever required.

CLASSES OF COMPUTERS

According to age and components:

- **Zeroth Generation**
  - Gears/Relays and Mechanical Systems
- **First Generation**
  - Vacuum Tubes and Monolithic Systems
- **Second Generation**
  - Transistors (Hardware) and Batch System (Software)
- **Third Generation**
  - Integrated Circuits (Hardware) and Multiprogramming Systems (Software)
- **Fourth Generation**
  - Microprocessor/VLSI and Personal Computing

CLASSES OF COMPUTERS

According to size and power:

- Portable Computers
- Microcomputer
- Minicomputer
- Supercomputer
- Mainframe

CLASSES OF COMPUTERS

According to design and application:

- General-purpose
- Special-purpose

CLASSES OF COMPUTERS

According to operation:

- Digital computers
- Hybrid computers
- Analog computers

CLASSES OF COMPUTERS

According to function:

- Server
- Workstation
- Terminal

CLASSES OF COMPUTERS

According to architecture:

- Uniprocessor
- Multiprocessor/ Parallel
- Multicomputer

COMPUTER TODAY?

- The first microcomputers became available in the 1970s and were used primarily by businesses. This lead to the personal computer revolution of the 1980s, in which microcomputers became a mainstream consumer product.

- As microcomputers grew in popularity, the name "microcomputer" faded and was replaced with other more specific terms.

- Minicomputers are digital computers, generally used in multi-user systems. They have high processing speed and high storage capacity than the microcomputers.

COMPUTER TODAY?

- A minicomputer is a type of computer that possesses most of the features and capabilities of a large computer but is smaller in physical size.

- Today, industry experts typically classify computers in seven categories: personal computers (desktop), mobile computers and mobile devices, game consoles, servers, mainframes, supercomputers, and embedded computers. A computer's size, speed, processing power, and price determine the category it best fits.

- This trend of computers and devices with technologies that overlap, called convergence, leads to computer manufacturers continually releasing newer models that include similar functionality and features.

PERSONAL COMPUTER

A **personal computer** is a computer that can perform all of its input, processing, output, and storage activities by itself. A personal computer contains a processor, memory, and one or more input, output, and storage devices. Personal computers also often contain a communications device.

✓ **Desktop Computer**

MOBILE COMPUTERS AND MOBILE DEVICES

A **mobile computer** is a personal computer you can carry from place to place. Similarly, a mobile device is a computing device small enough to hold in your hand. The most popular type of mobile computer is the notebook computer.

- ✓ **Notebook Computers**
- ✓ **Tablet PCs**
- ✓ **Mobile Devices**

GAME CONSOLE

A **game console** is a mobile computing device designed for single player or multiplayer video games. Standard game consoles use a handheld controller(s) as an input device(s); a television screen as an output device; and hard disks, optical discs, and/or memory cards for storage.

Three popular models are:

- **Microsoft's Xbox 360**
- **Nintendo's Wii (pronounced wee)**
- **Sony's PlayStation 3.**

SERVERS

- A **server** controls access to the hardware, software, and other resources on a network and provides a centralized storage area for programs, data, and information.
- **Servers** support from two to several thousand connected computers at the same time. People use personal computers or terminals to access data, information, and programs on a server.
- A **terminal** is a device with a monitor, keyboard, and memory.

MAINFRAMES

- A **mainframe** is a large, expensive, powerful computer that can handle hundreds or thousands of connected users simultaneously.
- **Mainframes** store huge amounts of data, instructions, and information. Most major corporations use mainframes for business activities.
- With **mainframes**, enterprises are able to bill millions of customers, prepare payroll for thousands of employees, and manage thousands of items in inventory.

SUPERCOMPUTERS

- A **supercomputer** is the fastest, most powerful computer — and the most expensive.
- The fastest **supercomputers** are capable of processing more than one quadrillion instructions in a single second.
- Applications requiring complex, sophisticated mathematical calculations use super computers. Large-scale simulations and applications in medicine, aerospace, automotive design, online banking, weather forecasting, nuclear energy research, and petroleum exploration use a supercomputer.

EMBEDDED COMPUTERS

- An **embedded computer** is a special-purpose computer that functions as a component in a larger product. A variety of everyday products contain embedded computers: Consumer electronics, such as;

    - **Home automation devices**
    - **Automobiles**
    - **Process controllers and robotics**
    - **Computer devices and office machines**

ADVANTAGES OF USING COMPUTERS

SPEED

When data, instructions, and information flow along electronic circuits in a computer, they travel at incredibly fast speeds. Many computers process billions or trillions of operations in a single second.

RELIABILITY

The electronic components in modern computers are dependable and reliable because they rarely break or fail.

CONSISTENCY

Given the same input and processes, a computer will produce the same results—consistently. Computers generate error-free results, provided the input is correct and the instructions work.

STORAGE

Computers store enormous amounts of data and make this data available for processing anytime it is needed.

COMMUNICATIONS

Most computers today can communicate with other computers, often wirelessly. Computers allow users to communicate with one another.

DISADVANTAGES OF USING COMPUTERS

VIOLATION PRIVACY

In many instances, where personal and confidential records stored on computers were not protected properly, individuals have found their privacy violated and identities stolen.

PUBLIC SAFETY

Adults, teens, and children around the world are using computers to share publicly their photos, videos, journals, music, and other personal information. Some of these unsuspecting, innocent computer users have fallen victim to crimes committed by dangerous strangers.

IMPACT ON LABOR FORCE

Although computers have improved productivity and created an entire industry with hundreds of thousands of new jobs, the skills of millions of employees have been replaced by computers. Thus, it is crucial that workers keep their education up-to-date. A separate impact on the labor force is that some companies are outsourcing jobs to foreign countries instead of keeping their homeland labor force employed.

DISADVANTAGES OF USING COMPUTERS

HEALTH RISKS

Prolonged or improper computer use can lead to health injuries or disorders. Computer users can protect themselves from health risks through proper workplace design, good posture while at the computer, and appropriately spaced work breaks. Two behavioral health risks are computer addiction and technology overload. Computer addiction occurs when someone becomes obsessed with using a computer. Individuals suffering from technology overload feel distressed when deprived of computers and mobile devices.

IMPACT ON ENVIRONMENT

Computer manufacturing processes and computer waste are depleting natural resources and polluting the environment.

Green computing involves reducing the electricity consumed and environmental waste generated when using a computer. Strategies that support green computing include recycling, regulating manufacturing processes, extending the life of computers, and immediately donating or properly disposing of replaced computers